

Practice Lab:

Install, configure and use
Maven

- This Practice Lab has no specific preceding Discussion Unit.
- In this Practice Lab we install, configure and use Maven, so that you can edit, compile and run Java client programs that target the DSE database server. As such, this lab expects an operating DSE cluster be accessible.
- Instructions in this practice lab are written for CentOS 7 using the command line, but should function for most Linux distros.
- Maven is an 8MB download, but any Java program compiles you perform will require reasonable Internet access.

Apache Maven: What is it ?

Apache Maven is "build automation software". Think yum(C), or apt-get(C), (iTunes), but for Java programs.

- Will create your project filesystem structure, compile, deploy, other.
- Will find, download, version, and manage all of your Java dependencies.
- And, do all of this in about 4 total commands.
- Supports Java, Scala, Ruby, C/C++ others



Apache Maven: How to install



Run all commands in this unit inside a Linux terminal window.

- Already installed ? `"mvn --version"`
- If not, Tar ball; unzip, place in PATH
 - <https://maven.apache.org/download.cgi>
 - <https://maven.apache.org/install.html>
 - Version 3.5.4, 8MB in size
 - Is a Java app, requires Java 8 (v 3.5.4)
 - `mvn --version`

Command 1: Initialize a new Java/DSE client project

- `cd /opt` (cd anywhere)
- `mvn archetype:generate`
 - DgroupId=`com.datastax.enablement.bootcamp`
 - DarchetypeArtifactId=maven-archetype-quickstart
 - DinteractiveMode=false
 - DartifactId=`my-app`

Will be the Java
Package name (as
generated)

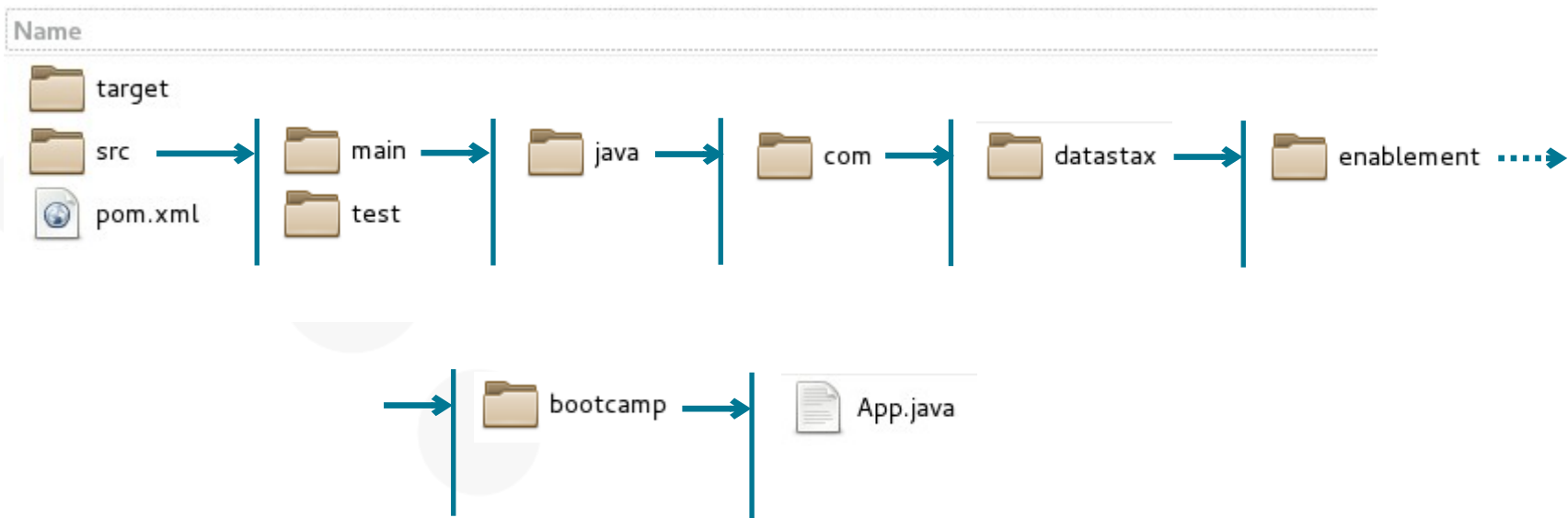


Project name (logical only)



Command 1: What just happened ?

-DgroupId=[com.datastax.enablement.bootcamp](#)



What is a pom.xml ?

- xml
- "project object model"
- Each project has one
- Tells Maven how to compile, etc, your project
- Edit once and done ?

```
<project xmlns="http://maven.apache.or ...  
    xsi:schemaLocation="http://mav ...  
    <modelVersion>4.0.0</modelVersion>  
  
    <groupId>com.datastax.enablement.bootcamp<  
/groupId>  
    <artifactId>my-app</artifactId>  
    <packaging>jar</packaging>  
    <version>1.0-SNAPSHOT</version>  
    <name>my-app</name>  
    <url>http://maven.apache.org</url>  
    <dependencies>  
        <dependency>  
            <groupId>junit</groupId>  
            <artifactId>junit</artifactId>  
            <version>3.8.1</version>  
            <scope>test</scope>  
        </dependency>  
    </dependencies>  
</project>
```

What's in App.java ?

```
package com.datastax.enablement.bootcamp;

/**
 * Hello world!
 *
 */
public class App
{
    public static void main( String[] args )
    {
        System.out.println( "Hello World!" );
    }
}
```

- Stub
- It runs (it's Hello World)
- Add DSE client code; structure

To compile and run the program-

- From the "project directory"

```
cd /opt/my-app
```

```
-DgroupId=com.datastax.enablement.bootcamp
```

```
-DartifactId=my-app
```

- # compiles

```
mvn package
```

- # generates the Java CLASSPATH

```
mvn dependency:build-classpath -Dmdep.outputFile=cp.txt
```

Left single
apostrophes

- # call to run the program (one line below)

```
java -cp target/my-app-1.0-SNAPSHOT.jar:`cat cp.txt`  
com.datastax.enablement.bootcamp.App
```



A POM that supports DSE/Java-

- Full/complete POM pasted below on Notes page
- But in effect, only need this/additional "dependency" block
- Jar files pulled, by default, from an online Maven repository
- Also pulls whatever Jars these Jars need, etc
- This repository can be (should be) internal for real/production apps

```
<dependencies>
  <dependency>
    <groupId>com.datastax.dse</groupId>
    <artifactId>dse-java-driver-core</artifactId>
    <version>1.6.7</version>
  </dependency>
</dependencies>
```

```
package com.datastax.enablement.bootcamp;
import com.datastax.driver.dse.DseCluster;
import com.datastax.driver.dse.DseSession;
import com.datastax.driver.core.Row;
public class App
{
    public static void main( String[] args )
    {
        DseCluster my_cluster = null;
        try {
            my_cluster = DseCluster.builder()
                .addContactPoint("127.0.0.1")
                .build();
            DseSession my_session = my_cluster.connect();
            Row my_row = my_session.execute("select * from system.local").one();
            System.out.println("DSE release version: " +
                my_row.getString("dse_version") );
        } finally {
            if (my_cluster != null) my_cluster.close();
        }
    }
}
```

An App.java that accesses DSE

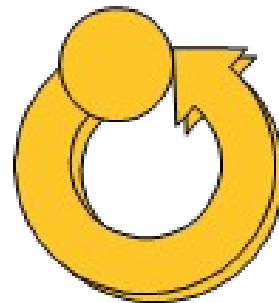
Next steps-

- Edit, compile, test, (repeat)

```
mvn package
```

```
mvn dependency:build-classpath -Dmdep.outputFile=cp.txt
```

```
java -cp target/my-app-1.0-SNAPSHOT.jar:`cat cp.txt`  
com.datastax.enablement.bootcamp.App
```



Practice Lab:

Lessons Learned



